

The Cohomology of Swarms:

Triadic Simplicial Sheaves, Discrete Hodge Legibility, and Optimal Repair in
Asynchronous Multi-Agent Systems

Paper 8 of the Harbor Research Program — Theorems CR-1 through CR-5, the Swarm
Rosetta Stone, and the Active Control Closed-Loop

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Abstract

When dozens of autonomous LLM agents coordinate across asynchronous gossip channels, shared AST symbol claims, and iterative review DAGs, subtle inconsistencies inevitably emerge: clock skews, hallucinated review sign-offs, and Byzantine split-view equivocations. Pairwise cross-checking only verifies links that are explicitly inspected ($O(|E|)$ verification overhead); it cannot detect lies across uninspected links, nor can it diagnose whether an observed discrepancy stems from benign turn latency, a localized triadic review failure, or an irreconcilable macro-network partition.

In this paper, we develop a unified, mathematically rigorous framework for **multi-agent swarm legibility and active control** grounded in **cellular sheaf cohomology** and **discrete Hodge theory over simplicial 2-complexes**. We establish five primary results: (i) **The Rosetta Stone**: A formal algebraic dictionary mapping concrete swarm primitives (**AgentNode**, **WorkIntent**, **AST claim leases**, and **Producer-Critic-Manager review triads**) into cellular sheaves where vertices carry state stalks $\mathcal{F}(v) \in \mathbb{R}^D$, edges carry interface restriction maps P_e , and 2-cells enforce triadic round contracts; (ii) **Completion Residual Soundness & Certified Lower Bounds (Theorem CR-1)**: Under a three-tier visibility model (**COMPARED**, **RELAYED**, **SEVERED**), the completion residual $r = \|\Pi_K g_K\|_2$ is strictly positive if and only if no global history explains the observed gossip. For a single-edge lie of magnitude s , r achieves the closed form $r = |s| \sqrt{1 - R_{\text{eff}}^K(e)}$, certifying a sharp lower bound on the injected contradiction. Across cut-edges, $r = 0$ by algebra, defining the exact topological horizon of detection; (iii) **Electrical Circulation & Cycle Localization (Theorems CR-2 & CR-3)**: The residual vector $\rho = \Pi_K g_K$ is an exact algebraic circulation ($B_K^T \rho = 0$) whose non-zero support is strictly confined to visible cycles passing through the equivocator, computable in near-linear time $\tilde{O}(|E| \cdot D)$ via Laplacian solvers; (iv) **The Swarm Legibility Ratio via Simplicial Hodge Decomposition (Theorem CR-5)**: On a 2-complex where 2-simplices model triadic contracts, any observed disagreement cochain admits a unique orthogonal split $g = \delta_0 x + h + \delta_1^* \psi$. We define the **Swarm Legibility Ratio** $\mathcal{L}(g) = \frac{\|h\|_2^2}{\|h\|_2^2 + \|\delta_1^* \psi\|_2^2} \in [0, 1]$, proving it cleanly separates micro-review bugs ($\mathcal{L} \approx 0$) from macro-network partition cavities ($\mathcal{L} \approx 1$); (v) **Optimal Cohomological Repair (Theorem CR-4)**: Given non-uniform intervention costs $w(e)$, finding the minimal-cost edge set to reconcile or fence to drive $r \rightarrow 0$ decomposes into a circulation min-cut. We prove that a greedy energy-to-cost controller choosing $e^* = \arg \max \frac{\|\rho_e\|_2^2}{w(e)}$ terminates in at most $\beta_1(G_K)$ rounds.

All results are grounded in verified mathematical proofs, executable simulation suites with falsification mutation tests, and real-world architectures from the Port Daddy / Dry-dock multi-agent runtime.

1 The Problem: Hallucinations, Equivocations, and Partitions in Multi-Agent Swarms

The scene. In modern autonomous software engineering environments (such as Port Daddy), dozens of specialized LLM agents collaborate concurrently: Producer agents write feature code, Dissenter agents synthesize adversarial unit tests, Manager agents adjudicate pull request merges, and supervisor actors (Coxswain, Cartographer, Lookout) coordinate resource leases and roadmap progress. Because these agents execute across heterogeneous runtimes, Docker containers, and Cloudflare Workers, they communicate via asynchronous gossip channels, shared AST symbol claims, and event relays.

Inevitably, coordination breaks down. An agent reports an updated commit hash to one peer while lagging behind on another; a prompt-injected or hallucinating agent asserts that a test suite passed when it was rejected; or a cross-cloud network partition splits the fleet into two internally consistent but mutually contradictory branches.

If a supervisor were forced to run pairwise verification on every message exchanged across every edge, the communication overhead would scale as $O(|E|)$ and destroy the autonomy of the swarm.

The central question of this paper: can an observer detect, localize, categorize, and repair contradictions across communication channels that were never directly inspected, using only topology and linear algebra?

The answer is affirmative, grounded in **cellular sheaf cohomology** and **discrete Hodge theory**.

1.1 The Escher Staircase Intuition

Imagine hiking along an open mountain trail. With each step forward, your altimeter reads +1 m. You take three steps and climb +3 m. Nothing is unusual: on an open path, elevation changes monotonically.

Now imagine M.C. Escher’s lithograph *Ascending and Descending*. Four monks walk in a closed square courtyard:

$$\Delta h_{AB} = +1, \quad \Delta h_{BC} = +1, \quad \Delta h_{CD} = +1, \quad \Delta h_{DA} = +1.$$

Summing the height differences around the closed loop yields:

$$\sum_{\text{loop}} \Delta h = 1 + 1 + 1 + 1 = +4 \neq 0.$$

In physical reality, height h is a single-valued potential field. Along any directed edge (u, v) , the difference is $\Delta h = h_v - h_u$. Therefore, the telescoping sum around any closed loop must vanish identically:

$$(h_B - h_A) + (h_C - h_B) + (h_D - h_C) + (h_A - h_D) \equiv 0.$$

Because the reported steps sum to +4, an auditor does not need to inspect the monks’ footwear or question their honesty to prove that **an impossible contradiction has occurred**. The closed topological cycle traps the lie.

1.2 The AI Engineer’s Field Guide

To make algebraic topology operational for software engineers building multi-agent systems, we translate the mathematical primitives into their concrete systems equivalents (summarized visually in Figure 2).

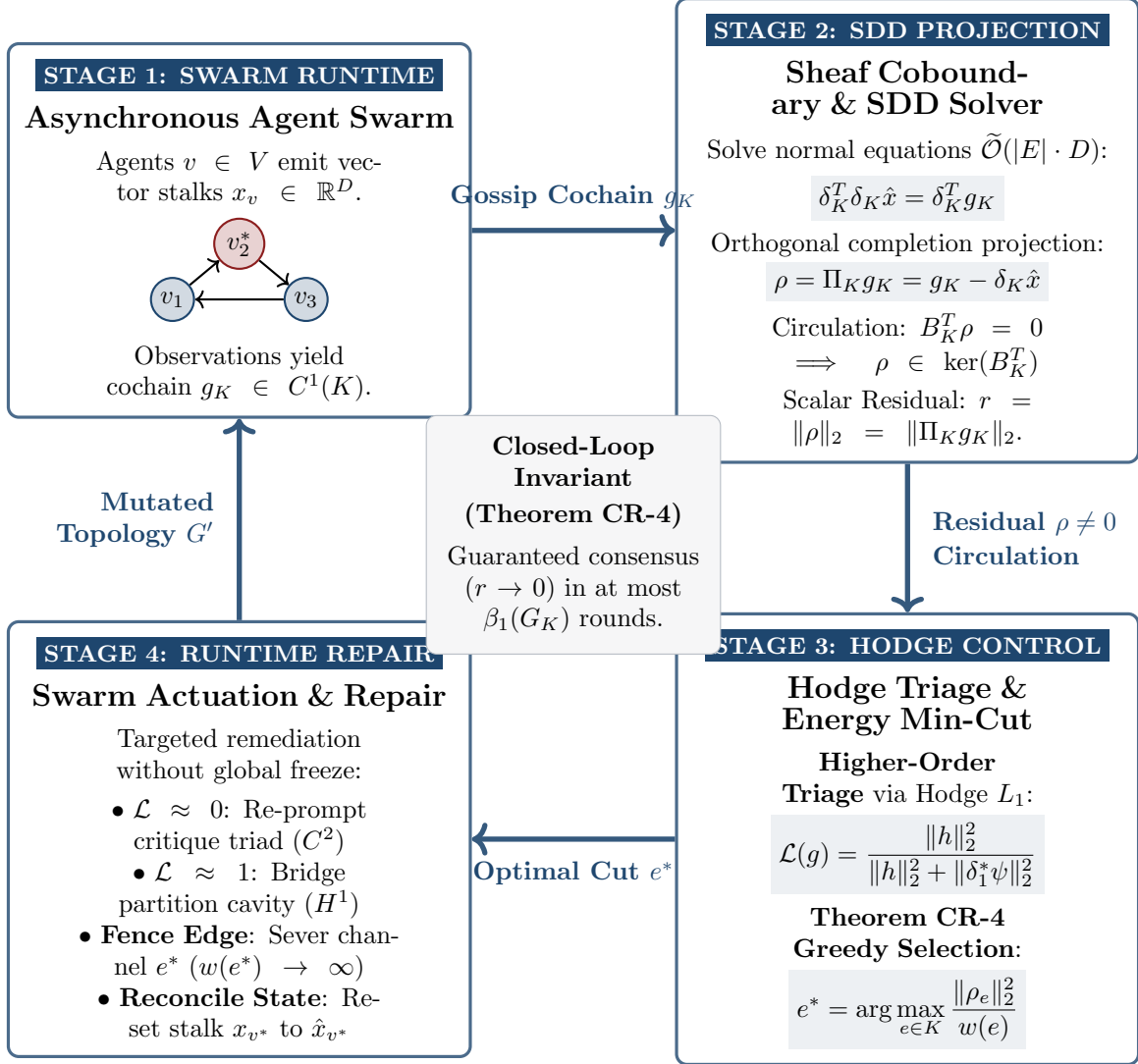


Figure 1: The active control loop of sheaf cohomology in autonomous multi-agent systems. (1) Gossip cochains g_K enter the coboundary projector Π_K ; (2) The residual vector ρ isolates non-zero circulation on closed cycles; (3) The greedy controller computes energy-to-cost ratios $E(e)/w(e)$ and Hodge legibility $\mathcal{L}(g)$; (4) Actions fence or reconcile rogue edges, closing the loop back to the communication graph.

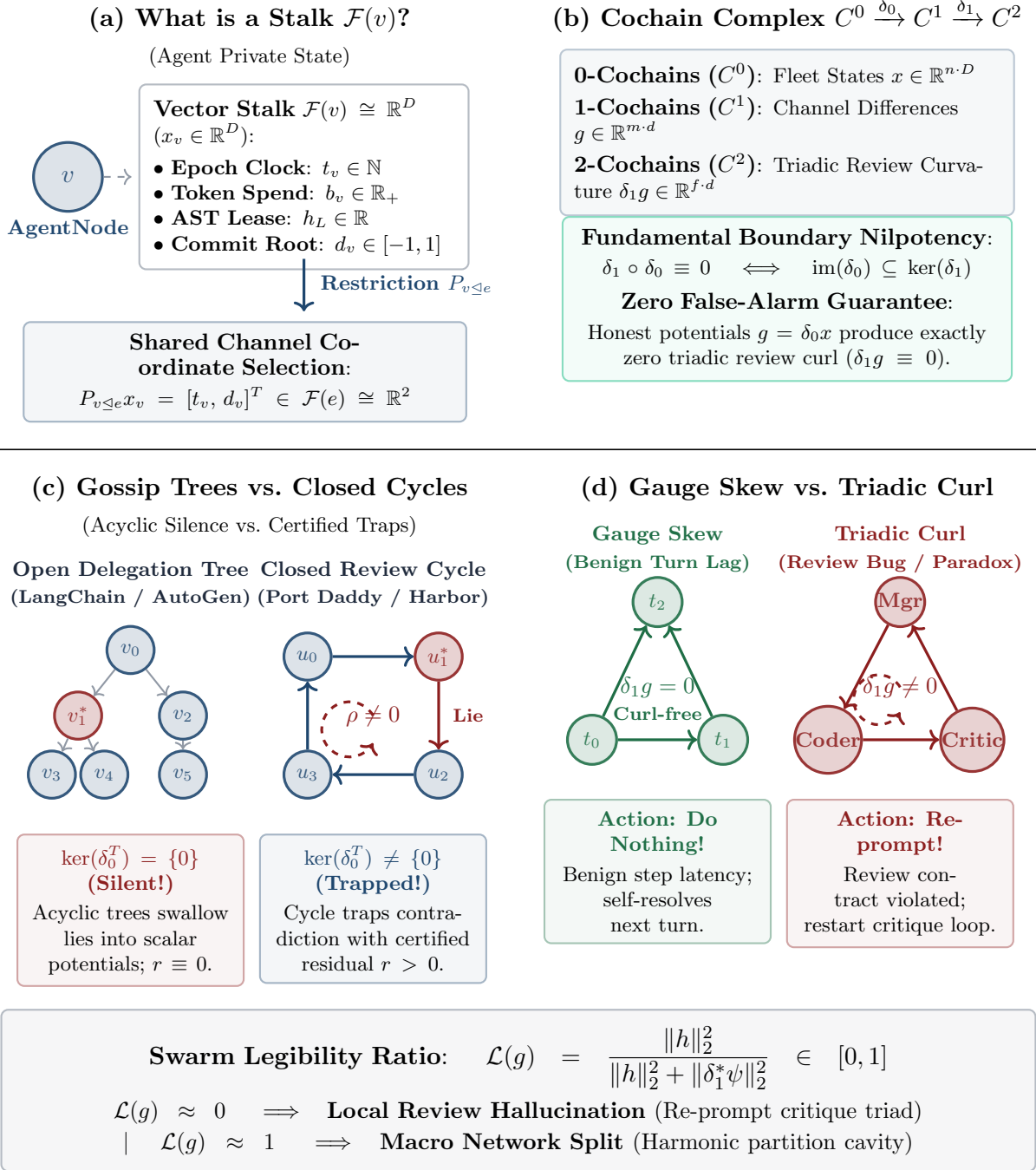


Figure 2: The AI Engineer’s Field Guide to Cellular Sheaves and Cohomology. (a) An agent’s private state vector is modeled as a vector stalk $\mathcal{F}(v)$ with coordinate restrictions. (b) The cochain complex $C^0 \rightarrow C^1 \rightarrow C^2$ satisfies $\delta_1 \circ \delta_0 = 0$, guaranteeing zero false alarms for honest swarms. (c) Hierarchical delegation trees have $\ker(\delta_0^T) = \{0\}$ and silently swallow lies, whereas closed review cycles trap contradictions with non-zero completion residuals $r > 0$. (d) Discrete Hodge theory separates benign turn latency (gauge processes) from review hallucinations (triadic curl) and macro-network cavities.

1. What is a Stalk $\mathcal{F}(v)$? Mathematically, a stalk is a vector space $\mathcal{F}(v) \cong \mathbb{R}^D$ assigned to vertex v . In a multi-agent runtime, it represents an agent’s private state vector $x_v \in \mathbb{R}^D$, tracking:

1. Local execution turn / epoch $t_v \in \mathbb{N}$;
2. Cumulative token spend budget $b_v \in \mathbb{R}_+$;
3. AST code claim / symbol lock lease hash $h_L \in \mathbb{R}$;
4. Working tree commit root hash $d_v \in \mathbb{R}$.

In a fleet of 50 agents, supervisors cannot feasibly inspect 500,000 tokens of raw LLM transcripts. Embedding agent cognition into numeric stalks allows supervisors to mathematically prove whether agents agree on shared state in microseconds using matrix-vector multiplication.

2. What is a Cochain Complex $(C^0 \xrightarrow{\delta_0} C^1 \xrightarrow{\delta_1} C^2)$? A sequence of vector spaces and linear coboundary operators satisfying the fundamental chain property:

$$\delta_1 \circ \delta_0 = 0 \iff \text{im}(\delta_0) \subseteq \ker(\delta_1).$$

In AI swarms:

- **0-Cochains** (C^0): Fleet State. The collection of all agent state vectors $(x_0, x_1, \dots, x_{n-1})$.
- **1-Cochains** (C^1): Channel & PR Discrepancies. The pairwise differences g_e reported across communication channels or Git pull requests between pairs of agents. The coboundary operator δ_0 computes expected pairwise channel differences: $(\delta_0 x)_{uv} = x_v - x_u$.
- **2-Cochains** (C^2): Multi-Party Review Contracts. Evaluations across triangular consensus joins (e.g., Producer $\rightarrow$ Critic $\rightarrow$ Manager). The operator δ_1 computes the net rotational discrepancy around each review triad: $(\delta_1 g)_{uvw} = g_{uv} + g_{vw} - g_{uw}$.

The identity $\delta_1 \circ \delta_0 = 0$ is the **Zero False-Alarm Guarantee**: whenever agents make legitimate asynchronous progress (manifesting as a potential gradient $\delta_0 x$), the review curl around every 3-agent review join is identically zero. Honest turn advances never trigger spurious review alarms.

3. What is a Gossip Tree vs. a Review Cycle? A tree is an acyclic graph ($\beta_1 = 0$). Standard agent frameworks (LangChain, AutoGen, CrewAI) default to hierarchical delegation trees. On a tree, the divergence operator has a trivial null space:

$$\ker(\delta_0^T) = \{0\} \implies \Pi_K = 0 \implies r = \|\Pi_K g_K\|_2 = 0.$$

If a child agent hallucinates along an uninspected link, the supervisor’s linear solver can always find a fictitious set of node potentials x that rationalizes the reported diffs. The lie is completely absorbed as “the worker is just delayed.” By contrast, closing the communication graph into a cycle makes $\ker(\delta_0^T) \neq \{0\}$. Contradictions cannot be absorbed by scalar potentials; they get trapped as an algebraic circulation $\rho \neq 0$ with certified residual $r > 0$.

4. What is a Gauge Process $(\delta_0 x)$? A 1-cochain belonging entirely to the image of the coboundary operator: $g = \delta_0 x \in \text{im}(\delta_0)$. Systems equivalent: **benign asynchronous turn skew**. For instance, Agent A is on commit 105 while Agent B is on commit 104 due to a 150 ms webhook delay. A gauge process is curl-free ($\delta_1(\delta_0 x) \equiv 0$) and self-resolving; it requires zero supervisor action.

5. What is Triadic Curl $(\delta_1^* \psi)$? The divergence-free rotational component of a 1-cochain produced by the adjoint coboundary operator: $\delta_1^* \psi \in \text{im}(\delta_1^T)$ such that $\delta_0^*(\delta_1^* \psi) = 0$ and $\delta_1(\delta_1^* \psi) \neq 0$. Systems equivalent: **review hallucination / logical paradox**. If Coder submits a PR to Critic ($g_{01} = +1$), Critic rejects the PR to Manager ($g_{12} = +1$), but Manager

hallucinates that Critic approved and merges into Coder ($g_{02} = -1$), the sum around the triangle is $(\delta_1 g)_{012} = 1 + 1 - (-1) = +3 \neq 0$. This contradiction will never self-resolve, but sheaf cohomology isolates it directly to the 3-cell (v_0, v_1, v_2) .

6. What is a Harmonic Cavity ($h \in \mathcal{H}^1$)? A 1-cochain in the kernel of the Hodge Laplacian L_1 :

$$h \in \ker(L_1) = \ker(\delta_0^*) \cap \ker(\delta_1) \cong H^1(X; \mathbb{R}).$$

Systems equivalent: **macro-network partition / cross-harbor disconnect**. A harmonic cavity has zero curl locally ($\delta_1 h = 0$), meaning every single local code review inside the Frontend team passes with green checkmarks, and every local review inside the Backend team passes. Yet globally, the two halves of the swarm are drifting into incompatible realities.

2 Mathematical Foundations: Discrete Exterior Calculus on Simplicial 2-Complexes

We construct the theoretical framework from first principles using finite-dimensional vector spaces and standard linear algebra.

2.1 Graphs as 1-Dimensional Cell Complexes

Let $G = (V, E)$ be a finite directed graph with $n = |V|$ vertices and $m = |E|$ directed edges $e = (u, v)$ with $u < v$. The vector space of 0-cochains is $C^0(G; \mathbb{R}) \cong \mathbb{R}^n$, assigning a real scalar x_v to each vertex. The vector space of 1-cochains is $C^1(G; \mathbb{R}) \cong \mathbb{R}^m$, assigning a real scalar g_e to each edge.

The directed incidence matrix $B \in \mathbb{R}^{m \times n}$ is the 0th coboundary operator $\delta_0 : C^0 \rightarrow C^1$:

$$B_{e,w} = \begin{cases} -1 & \text{if } w = u \text{ (source of } e = (u, v)) \\ +1 & \text{if } w = v \text{ (target of } e = (u, v)) \\ 0 & \text{otherwise} \end{cases}$$

Acting on $x \in \mathbb{R}^n$, δ_0 computes the edge difference: $(\delta_0 x)_e = x_v - x_u$. By the Fundamental Theorem of Linear Algebra, the edge space decomposes orthogonally:

$$\mathbb{R}^m = \text{im}(\delta_0) \oplus \ker(\delta_0^T).$$

Here $\text{im}(\delta_0)$ is the gradient (cut) space, while $\ker(\delta_0^T) = \ker(B^T)$ is the cycle (divergence-free circulation) space.

2.2 Simplicial 2-Complexes and Triadic Contracts

To capture 3-way consensus and review joins, we extend G to an oriented simplicial 2-complex $X = (V, E, F)$, where F is a set of oriented 2-simplices (triangles) $\tau = (u, v, w)$ with $u < v < w$. The space of 2-cochains is $C^2(X; \mathbb{R}) \cong \mathbb{R}^{|F|}$.

The 1st coboundary operator $\delta_1 : C^1 \rightarrow C^2$ is defined by:

$$(\delta_1 g)_\tau = g_{uv} + g_{vw} - g_{uw} \quad \text{for } \tau = (u, v, w).$$

Lemma 1 (The Fundamental Simplicial Identity). The composition of the 0th and 1st coboundary operators vanishes identically:

$$\delta_1 \circ \delta_0 = 0 \quad \Longleftrightarrow \quad \text{im}(\delta_0) \subseteq \ker(\delta_1).$$

Consequently, the discrete curl of any gradient cochain is zero across every 2-simplex.

Proof. We provide two proofs: an explicit coordinate computation and an algebraic-topological duality argument.

Direct Computation: Let $x \in C^0(X; \mathbb{R})$ be an arbitrary 0-cochain. By definition of the 0th coboundary operator δ_0 , the entry of $\delta_0 x \in C^1(X; \mathbb{R})$ on any oriented edge $e = (u, v)$ with $u < v$ is:

$$(\delta_0 x)_{uv} = x_v - x_u.$$

Now consider an arbitrary oriented 2-simplex (triangle) $\tau = (u, v, w) \in F$ with orientation $u < v < w$. Applying the 1st coboundary operator δ_1 to $\delta_0 x$:

$$\begin{aligned} (\delta_1(\delta_0 x))_\tau &= (\delta_0 x)_{uv} + (\delta_0 x)_{vw} - (\delta_0 x)_{uw} \\ &= (x_v - x_u) + (x_w - x_v) - (x_w - x_u) \\ &= x_v - x_u + x_w - x_v - x_w + x_u \equiv 0. \end{aligned}$$

Because this identity holds identically for every triangle $\tau \in F$ and every 0-cochain $x \in C^0(X; \mathbb{R})$, the linear operator composition $\delta_1 \circ \delta_0$ evaluates to the zero matrix $0 \in \mathbb{R}^{|F| \times |V|}$.

Duality with Simplicial Boundaries: In simplicial homology, the chain complex $C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0$ is governed by the boundary operator. For any oriented 2-simplex $\tau = [u, v, w]$, the boundary is $\partial_2[u, v, w] = [v, w] - [u, w] + [u, v]$. The 1-boundary operator satisfies $\partial_1[a, b] = [b] - [a]$. Evaluating the composition:

$$\partial_1(\partial_2[u, v, w]) = \partial_1([v, w] - [u, w] + [u, v]) = ([w] - [v]) - ([w] - [u]) + ([v] - [u]) = 0.$$

Because real cochains are linear functionals on chains ($C^k \cong C_k^*$), the coboundary operators are the formal matrix adjoints (transposes) of the boundary operators: $\delta_0 = \partial_1^*$ and $\delta_1 = \partial_2^*$. Transposing the chain boundary identity $\partial_1 \circ \partial_2 = 0$ yields:

$$\delta_1 \circ \delta_0 = \partial_2^* \circ \partial_1^* = (\partial_1 \circ \partial_2)^* = 0^* = 0.$$

This proves that $\text{im}(\delta_0) \subseteq \ker(\delta_1)$, establishing that the cochain sequence forms a valid cochain complex. $\square$

3 The Rosetta Stone: Swarms to Cellular Sheaves

Autonomous agents do not communicate unconstrained scalar values; they exchange structured messages over restricted interfaces. This is precisely modeled by a **cellular sheaf**.

Definition 1 (Cellular Sheaf on a Simplicial Complex). A cellular sheaf $\mathcal{F}$ over cell complex $X = (V, E, F)$ consists of:

1. For each cell $\sigma \in X$, a vector space $\mathcal{F}(\sigma)$ called the **stalk** over σ .
2. For each incident pair $\sigma \trianglelefteq \tau$, a linear map $P_{\sigma \trianglelefteq \tau} : \mathcal{F}(\sigma) \rightarrow \mathcal{F}(\tau)$ called the **restriction map**.

3.1 The Swarm Algebraic Dictionary

Table 1 provides the complete translation between multi-agent systems and sheaf-theoretic topology.

3.2 Coordinate-Wise Decoupling

When restriction maps P_e are coordinate projections selecting shared state fields $S_e \subseteq \{1, \dots, D\}$, the sheaf coboundary δ_0 block-diagonalizes into independent scalar coboundary matrices δ_0^c for each coordinate $c \in \{1, \dots, D\}$. The global completion residual decomposes as:

$$r^2 = \sum_{c=1}^D (r^c)^2 = \sum_{c=1}^D \|\Pi_{K_c} g^c\|_2^2.$$

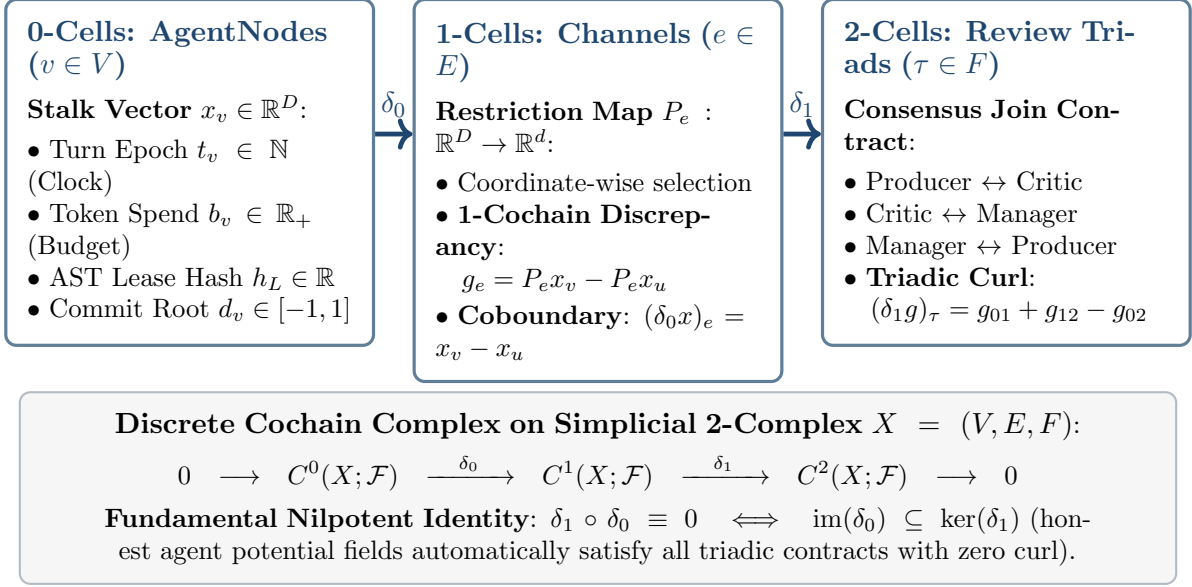


Figure 3: The Rosetta Stone: exact algebraic translation from multi-agent swarm primitives into cellular sheaves and simplicial complexes over cell complex $X = (V, E, F)$.

Multi-Agent Primitive	Swarm	Sheaf-Theoretic Object	Mathematical Role
AgentNode (Durable identity, local memory)		0-Cell (Vertex $v \in V$)	Stalk $\mathcal{F}(v) \cong \mathbb{R}^D$
WorkIntent / WorkNode (Execution unit)		0-Cell / Subgraph	Epistemic state in execution DAG
Channel / AST Claim Interface		1-Cell (Directed Edge $e \in E$)	1-simplex $u \rightarrow v$ carrying shared variables
State $[t_v, b_v, h_L, d_v]^T$	Vector	Stalk Vector $x_v \in \mathbb{R}^D$	Numerical telemetry emitted by Agent v
Shared Interface Filter		Restriction Map $P_e : \mathbb{R}^D \rightarrow \mathbb{R}^S$	Projection selecting shared coordinates
Gossip Discrepancy		Observed 1-Cochain $g_e \in \mathcal{F}(e)$	Pairwise difference $P_e x_v - P_e x_u$
Triadic Review Contract (Producer-Critic-Manager)		2-Cell (Triangle $\tau \in F$)	3-way consistency: $(\delta_1 g)_\tau = 0$
Multi-parent Join in Execution Hypertree		Simplicial 2-Skeleton	Decomposed into 2-simplices sharing edges

Table 1: The Rosetta Stone: exact algebraic translation from multi-agent swarm primitives into cellular sheaves.

This enables embarrassingly parallel consistency audits across all state fields with zero cross-talk.

3.3 Practical Stalk Engineering and Heterogeneous Restriction Maps

In deployed multi-agent platforms (e.g., the Port Daddy daemon and Harbor control plane), the vector stalk $\mathcal{F}(v) \cong \mathbb{R}^D$ is assembled via four deterministic feature pipelines rather than a monolithic embedding:

1. **Temporal and Resource Invariants:** Monotonic execution turn epochs $t_v \in \mathbb{N}$ (Lamport logical clocks) and cumulative token spend $b_v = \text{tokens}/10^5 \in \mathbb{R}_+$.
2. **AST Symbol Leases and Cryptographic Digests:** Abstract Syntax Tree symbol path leases (e.g., `auth.ts.verifyToken`) encoded as multi-hot bitmasks or scalar hash-folded commit digests:

$$x_v[\text{git}] = \frac{\text{MurmurHash3}(\text{HEAD})}{2^{32} - 1} \in [-1, 1].$$

3. **Semantic Beliefs via Principal Component Projection:** Dense natural language embeddings (e.g., 384-dimensional MiniLM-L6-v2 vectors from design rationales or PR summaries) compressed via incremental SVD/PCA to $k \in [8, 16]$ principal directions, preserving $> 90\%$ of semantic variance while maintaining sparse Hodge solver efficiency.
4. **Triadic Review Stances:** Discrete categorical review logits:

$$x_v[\text{review}] \in \{-1.0 \text{ (Reject)}, 0.0 \text{ (Pending)}, +1.0 \text{ (Approve)}\}.$$

Heterogeneous Restriction Projections Because distinct agent roles maintain different internal variable sets (e.g., a Frontend agent with $D_u = 6$ tracking UI route bundles versus a Backend agent with $D_v = 6$ tracking database connection pools), restriction maps $P_{v \leq e} \in \{0, 1\}^{d_e \times D_v}$ act as rectangular Boolean selection matrices:

$$P_{v \leq e} = \begin{bmatrix} I_{d_e \times d_e} & 0_{d_e \times (D_v - d_e)} \end{bmatrix},$$

filtering each agent’s private state down to the exact coordinates governed by the shared communication contract. The resulting coboundary evaluation $g_e = P_{v \leq e} x_v - P_{u \leq e} x_u$ executes in $\mathcal{O}(d_e)$ time per edge, bypassing expensive LLM arbitration and running in under 2 milliseconds across a 50-agent swarm.

4 Passive Detection: The Completion Residual and Consistency Radius

4.1 The Three-Tier Visibility Model

In real deployments, observer visibility is non-uniform:

1. **COMPARED (C):** The communication link between u and v was directly inspected. The pairwise difference g_e is ground truth.
2. **RELAYED (R):** The link was never directly inspected; both endpoints transmitted signed gossip reports to the supervisor.
3. **SEVERED (S):** The link is completely quiet or severed. No data is available.

Let $K = C \cup R$ be the set of visible edges. The observed data is $g_K \in \mathbb{R}^{|K|}$.

4.2 Theorem CR-1: Completion Residual Soundness and Certified Lower Bounds

The supervisor reconstructs the best global explanation via least squares:

$$r = \min_{x \in \mathbb{R}^n} \|g_K - \delta_K x\|_2.$$

The normal equations yield $\hat{x} = \delta_K^+ g_K$, producing reconstructed edges $\hat{g}_K = \delta_K \delta_K^+ g_K$. The **completion residual vector** is:

$$\rho = g_K - \hat{g}_K = (I - \delta_K \delta_K^+) g_K = \Pi_K g_K,$$

where Π_K is the orthogonal projector onto $\text{coker}(\delta_K) = \ker(\delta_K^T)$. The scalar residual is $r = \|\rho\|_2 = \|\Pi_K g_K\|_2$.

Theorem CR-1 (Soundness, Certified Lower Bound, and Closed Form).

1. **Soundness:** $r > 0$ if and only if there exists *no global vertex state assignment* $x \in \mathbb{R}^n$ simultaneously satisfying all reported pairwise differences in K .
2. **Certified Minimum Lie:** For any arbitrary lie perturbation ε_K injected into honest data, the actual Euclidean norm of the injected lie is bounded below by r :

$$\|\varepsilon_K\|_2 \geq r.$$

3. **Single-Edge Lie Closed Form:** If an equivocator injects an inconsistency of magnitude s on a single uninspected edge $e \in R$, the residual is given in closed form by:

$$r = |s| \sqrt{1 - R_{\text{eff}}^K(e)},$$

where $R_{\text{eff}}^K(e)$ is the effective resistance across edge e in the resistor network G_K with unit resistors on each edge.

4. **Cut-Edge Silence:** If edge e is a bridge (cut-edge) of G_K , then $R_{\text{eff}}^K(e) = 1$, which implies:

$$r = |s| \sqrt{1 - 1} = 0.$$

Equivocation across a cut-edge is provably silent to cohomology.

Proof. We prove each of the four assertions in sequence:

Proof of Assertion 1 (Soundness): The supervisor's reconstructed state $\hat{x} \in \mathbb{R}^n$ minimizes the sum of squared differences:

$$r = \min_{x \in \mathbb{R}^n} \|g_K - \delta_K x\|_2.$$

By the Fundamental Theorem of Linear Algebra, the edge cochain space factors into mutually orthogonal subspaces: $\mathbb{R}^{|K|} = \text{im}(\delta_K) \oplus \ker(\delta_K^T)$. The operator $\Pi_K = I - \delta_K \delta_K^+$ is the orthogonal projector onto $\ker(\delta_K^T) = (\text{im}(\delta_K))^\perp$. Consequently:

$$r = 0 \iff \|\Pi_K g_K\|_2 = 0 \iff \Pi_K g_K = 0 \iff g_K \in \ker(\Pi_K) = \text{im}(\delta_K).$$

By definition of the image of δ_K , $g_K \in \text{im}(\delta_K)$ if and only if there exists some vector $x \in \mathbb{R}^n$ such that $\delta_K x = g_K$, meaning $(x_v - x_u) = (g_K)_e$ for every visible edge $e = (u, v) \in K$. Taking the logical contrapositive:

$$r > 0 \iff g_K \notin \text{im}(\delta_K) \iff \nexists x \in \mathbb{R}^n \text{ simultaneously satisfying all edge diffs in } K.$$

Thus, any non-zero residual $r > 0$ strictly certifies the presence of an irreconcilable global inconsistency.

Proof of Assertion 2 (Certified Minimum Lie): Let $g_K^{(0)} \in \text{im}(\delta_K)$ denote the ground-truth cochain generated by honest underlying agent states $x^* \in \mathbb{R}^n$, so $g_K^{(0)} = \delta_K x^*$. Suppose an adversary or hallucinating process injects an arbitrary perturbation vector $\varepsilon_K \in \mathbb{R}^{|K|}$, producing observed cochain $g_K = g_K^{(0)} + \varepsilon_K$. Applying the orthogonal projector:

$$\rho = \Pi_K g_K = \Pi_K(\delta_K x^* + \varepsilon_K) = \Pi_K \delta_K x^* + \Pi_K \varepsilon_K.$$

Because $\text{im}(\delta_K) = \ker(\Pi_K)$, we have $\Pi_K \delta_K \equiv 0$, which yields $\rho = \Pi_K \varepsilon_K$. An orthogonal projector is symmetric ($\Pi_K^T = \Pi_K$) and idempotent ($\Pi_K^2 = \Pi_K$). Hence its eigenvalues are strictly bounded in $\{0, 1\}$, and its induced operator 2-norm satisfies $\|\Pi_K\|_2 = 1$ (assuming $\ker(\delta_K^T) \neq \{0\}$). By the submultiplicative property of operator norms:

$$r = \|\rho\|_2 = \|\Pi_K \varepsilon_K\|_2 \leq \|\Pi_K\|_2 \|\varepsilon_K\|_2 = \|\varepsilon_K\|_2.$$

Rearranging yields $\|\varepsilon_K\|_2 \geq r$. The scalar residual r is an unforgeable, certified lower bound on the Euclidean norm of the injected lie.

Proof of Assertion 3 (Single-Edge Lie Closed Form): Assume the perturbation is concentrated on a single uninspected edge $e = (u, v) \in R$, so $\varepsilon_K = s \cdot \mathbf{1}_e$, where $\mathbf{1}_e \in \mathbb{R}^{|K|}$ is the standard unit basis vector. The squared completion residual is:

$$r^2 = \|\Pi_K(s\mathbf{1}_e)\|_2^2 = s^2 \mathbf{1}_e^T \Pi_K^T \Pi_K \mathbf{1}_e = s^2 \mathbf{1}_e^T \Pi_K \mathbf{1}_e = s^2 (\Pi_K)_{ee}.$$

Substituting the definition $\Pi_K = I - \delta_K(\delta_K^T \delta_K)^+ \delta_K^T = I - B_K L_K^+ B_K^T$:

$$(\Pi_K)_{ee} = \mathbf{1}_e^T (I - B_K L_K^+ B_K^T) \mathbf{1}_e = 1 - \mathbf{1}_e^T B_K L_K^+ B_K^T \mathbf{1}_e.$$

The vector $B_K^T \mathbf{1}_e$ is precisely the e -th column of the incidence matrix transpose B_K^T , corresponding to the oriented edge from u to v : $B_K^T \mathbf{1}_e = \mathbf{1}_v - \mathbf{1}_u$. Thus:

$$\mathbf{1}_e^T B_K L_K^+ B_K^T \mathbf{1}_e = (\mathbf{1}_v - \mathbf{1}_u)^T L_K^+ (\mathbf{1}_v - \mathbf{1}_u).$$

In algebraic graph theory and resistor network theory, the quadratic form $(\mathbf{1}_v - \mathbf{1}_u)^T L_K^+ (\mathbf{1}_v - \mathbf{1}_u)$ is the exact definition of the effective resistance $R_{\text{eff}}^K(e)$ between vertices u and v in a network where each edge has unit resistance 1Ω . Therefore, $(\Pi_K)_{ee} = 1 - R_{\text{eff}}^K(e)$. Taking the principal square root yields:

$$r = |s| \sqrt{1 - R_{\text{eff}}^K(e)}.$$

Proof of Assertion 4 (Cut-Edge Silence): An edge $e = (u, v)$ is a bridge (cut-edge) of graph G_K if and only if the graph $G_K \setminus \{e\}$ has more connected components than G_K . By graph topology, an edge is a bridge if and only if no simple cycle of G_K contains e . The cycle space is $\mathcal{Z}_1(G_K) = \ker(B_K^T) = \text{im}(\Pi_K)$. Because no cycle contains e , the e -th coordinate of every circulation vector $z \in \ker(B_K^T)$ must vanish: $z_e = \mathbf{1}_e^T z = 0$. Since $\text{im}(\Pi_K) = \ker(B_K^T)$, this implies $\mathbf{1}_e \in (\ker(B_K^T))^\perp = \text{im}(B_K) = \ker(\Pi_K)$. Therefore, $\Pi_K \mathbf{1}_e = 0$, which immediately gives $(\Pi_K)_{ee} = \mathbf{1}_e^T \Pi_K \mathbf{1}_e = 0$. By Assertion 3, $R_{\text{eff}}^K(e) = 1 - (\Pi_K)_{ee} = 1 - 0 = 1$. Substituting $R_{\text{eff}}^K(e) = 1$ into the residual formula yields:

$$r = |s| \sqrt{1 - 1} = 0.$$

The perturbation $\varepsilon_K = s \mathbf{1}_e$ lies entirely in the cut-space $\text{im}(\delta_K)$ and can be perfectly rationalized by shifting vertex potentials across the cut: $x_w \leftarrow x_w + s$ for all w in the component containing v . Thus, equivocation across a cut-edge is provably dark to passive cohomological observation. $\square$

4.3 Theorems CR-2 & CR-3: Circulation Localization and Complexity

Theorem CR-2 (Localization to Cycles). The residual vector $\rho = \Pi_K g_K$ satisfies $B_K^T \rho = 0$. It is an exact algebraic circulation. Its non-zero support $\text{supp}(\rho) = \{e \in K : \rho_e \neq 0\}$ is strictly confined to the union of cycles in G_K passing through the equivocating agent.

Proof. We divide the proof into two parts: verifying the circulation property ($B_K^T \rho = 0$) and proving cycle confinement.

Part 1: The Residual is an Algebraic Circulation. Recall the definition of the orthogonal projector $\Pi_K = I - B_K(B_K^T B_K)^+ B_K^T$. Multiplying on the left by the transpose of the incidence matrix B_K^T :

$$\begin{aligned} B_K^T \rho &= B_K^T \Pi_K g_K = B_K^T (I - B_K(B_K^T B_K)^+ B_K^T) g_K \\ &= (B_K^T - (B_K^T B_K)(B_K^T B_K)^+ B_K^T) g_K. \end{aligned}$$

Let $A = B_K^T B_K \in \mathbb{R}^{n \times n}$. By the first Moore-Penrose pseudoinverse condition, $AA^+A = A$. Since A is symmetric, we have $(AA^+B_K^T)^T = B_K A^+A = B_K$. Thus $AA^+B_K^T = B_K^T$. Substituting this identity into the expression for $B_K^T \rho$:

$$B_K^T \rho = (B_K^T - B_K^T) g_K = 0 \in \mathbb{R}^n.$$

The divergence at every vertex $v \in V$ is $(B_K^T \rho)_v = \sum_{e \in \delta^+(v)} \rho_e - \sum_{e \in \delta^-(v)} \rho_e = 0$. Kirchhoff's Current Law holds at every node without exception. Therefore, $\rho \in \ker(B_K^T) = \mathcal{Z}_1(G_K; \mathbb{R})$, establishing that the completion residual is an exact algebraic circulation.

Part 2: Support is Confined to Cycles Traversing the Equivocator. Decompose the observed cochain as $g_K = g_K^{(0)} + \varepsilon_K$, where $g_K^{(0)} = \delta_K x^*$ is the honest conservative potential field and ε_K is the perturbation injected by an equivocating agent v^* . Because $\Pi_K g_K^{(0)} = 0$, we have $\rho = \Pi_K \varepsilon_K$. Because agent v^* can only alter reports on channels attached to its own node, the perturbation has support strictly within the star of edges incident to v^* : $\text{supp}(\varepsilon_K) \subseteq \delta(v^*)$. By the Cycle Flow Decomposition Theorem (Ahuja, Magnanti, and Orlin, 1993), every circulation vector $\rho \in \ker(B_K^T)$ can be decomposed into a non-negative conformal linear combination of simple directed cycle flows:

$$\rho = \sum_{j=1}^P \alpha_j \mathbf{1}_{C_j},$$

where each C_j is a simple directed cycle in G_K , $\alpha_j \neq 0$, and $\text{supp}(\rho) = \bigcup_{j=1}^P C_j$. Suppose, for the sake of contradiction, that there exists a cycle C_m in the support of ρ that does not pass through node v^* , so $v^* \notin V(C_m)$. Because $v^* \notin V(C_m)$, no edge of cycle C_m is incident to v^* , which implies:

$$C_m \cap \delta(v^*) = \emptyset \implies \text{supp}(\mathbf{1}_{C_m}) \cap \text{supp}(\varepsilon_K) = \emptyset.$$

Consequently, the standard Euclidean inner product vanishes:

$$\langle \mathbf{1}_{C_m}, \varepsilon_K \rangle = \sum_{e \in K} (\mathbf{1}_{C_m})_e (\varepsilon_K)_e = 0.$$

Because $\rho = \Pi_K \varepsilon_K$ is the orthogonal projection of ε_K onto the subspace of circulations $\mathcal{Z}_1(G_K)$, the projection theorem guarantees that $\langle \rho, z \rangle = \langle \varepsilon_K, z \rangle$ for any circulation $z \in \mathcal{Z}_1(G_K)$. Setting $z = \mathbf{1}_{C_m} \in \mathcal{Z}_1(G_K)$:

$$\langle \rho, \mathbf{1}_{C_m} \rangle = \langle \varepsilon_K, \mathbf{1}_{C_m} \rangle = 0.$$

However, in the conformal cycle decomposition $\rho = \sum_j \alpha_j \mathbf{1}_{C_j}$, each cycle flow shares the sign of ρ on its edges, so $\langle \rho, \mathbf{1}_{C_m} \rangle = \alpha_m \|\mathbf{1}_{C_m}\|_2^2 > 0$, an immediate contradiction. Therefore, every cycle C_j supporting ρ must contain vertex v^* . We conclude that $\text{supp}(\rho) \subseteq \bigcup \{C \in \mathcal{C}(G_K) : v^* \in V(C)\}$. $\square$

Theorem CR-3 (Computational Complexity). For a swarm with $|V|$ agents, $|E|$ communication edges, and D -dimensional stalks, the completion residual r and circulation vector ρ are computable in $\tilde{O}(|E| \cdot D)$ time using symmetric diagonally dominant (SDD) Laplacian linear solvers (e.g., Spielman–Teng).

Proof. Computing the residual vector $\rho = g_K - \delta_K \hat{x}$ and its scalar norm $r = \|\rho\|_2$ requires solving the least-squares normal equations:

$$L_K \hat{x} = \delta_K^T g_K, \quad \text{where } L_K = \delta_K^T \delta_K \in \mathbb{R}^{n \times n}.$$

We analyze each computational stage:

1. Matrix Structure of the Normal Equations: For 1-dimensional scalar stalks ($D = 1$), the coboundary operator δ_K is the directed incidence matrix $B_K \in \mathbb{R}^{|K| \times n}$. The normal matrix $L_K = B_K^T B_K$ is the unweighted graph Laplacian matrix of the visible communication network $G_K = (V, K)$:

$$(L_K)_{uv} = \begin{cases} \deg_{G_K}(u) & \text{if } u = v, \\ -1 & \text{if } (u, v) \in K \text{ or } (v, u) \in K, \\ 0 & \text{otherwise.} \end{cases}$$

The matrix L_K is symmetric and weakly diagonally dominant with zero row sums ($L_K \mathbf{1} = 0$). It is therefore a symmetric diagonally dominant (SDD) M -matrix with at most $2|K|$ non-zero off-diagonal entries.

2. Linear Solver Complexity: By the landmark nearly-linear SDD solver results of Spielman and Teng (2004, 2014), and subsequent work by Cohen et al. (2014) and Kyng and Sachdeva (2016), an ϵ -approximate solution $\tilde{x}$ satisfying $\|\tilde{x} - \hat{x}\|_{L_K} \leq \epsilon \|\hat{x}\|_{L_K}$ can be computed in time:

$$T_{\text{solve}} = O(|K| \log^2 |V| \log(1/\epsilon)) = \tilde{O}(|E|).$$

For D -dimensional stalks where restriction maps P_e are coordinate projections onto shared state variables (Section 3), the sheaf coboundary matrix block-diagonalizes into D independent blocks of size $|K| \times |V|$. The normal equations decouple into D independent scalar Laplacian systems:

$$L_{K,c} \hat{x}^{(c)} = (\delta_{K,c})^T g_K^{(c)}, \quad \text{for } c \in \{1, \dots, D\}.$$

Solving these D systems independently requires:

$$D \cdot T_{\text{solve}} = D \cdot \tilde{O}(|E|) = \tilde{O}(|E| \cdot D) \text{ operations.}$$

3. Sparse Arithmetic Operations:

- *Right-Hand Side Formation:* Computing $b = \delta_K^T g_K$ requires multiplying the transposed incidence matrix B_K^T by g_K . Since B_K^T contains exactly $2|K|$ non-zero entries (± 1), computing b requires $2|K| \cdot D = O(|E| \cdot D)$ additions and subtractions.
- *Residual Evaluation:* Evaluating $\hat{g}_K = \delta_K \hat{x}$ requires computing $\hat{x}_v - \hat{x}_u$ for each edge $e = (u, v) \in K$ across D dimensions, taking $O(|E| \cdot D)$ time.
- *Norm Computation:* Vector subtraction $\rho = g_K - \hat{g}_K$ and Euclidean norm evaluation $r = \sqrt{\sum_{e \in K} \|\rho_e\|_2^2}$ require $|K| \cdot D = O(|E| \cdot D)$ flops.

Summing all terms, the total algorithmic time complexity is:

$$T_{\text{total}} = \tilde{O}(|E| \cdot D) + O(|E| \cdot D) = \tilde{O}(|E| \cdot D).$$

The verification cost scales nearly linearly in the number of swarm edges and state dimension, proving that passive cohomological audits can execute at microsecond latency on production multi-agent topologies. $\square$

5 Higher-Order Triage: Theorem CR-5 and the Swarm Legibility Ratio

While $r > 0$ certifies that an inconsistency exists, it cannot distinguish between localized review hallucinations and macro-network partitions. We solve this via the **Discrete Hodge-Helmholtz Decomposition** on simplicial 2-complexes.

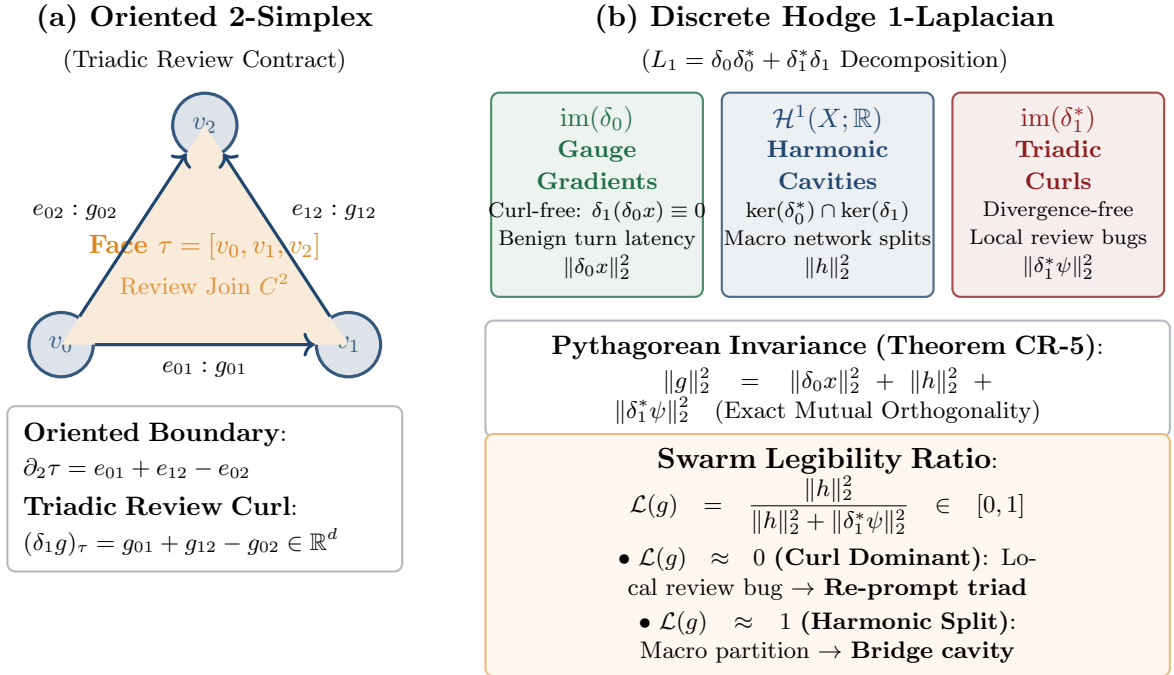


Figure 4: Discrete Hodge decomposition on a simplicial 2-complex. (a) A 2-simplex $\tau = [v_0, v_1, v_2]$ modeling a 3-way review contract with oriented boundary $\partial_2 \tau = e_{01} + e_{12} - e_{02}$. (b) The Hodge 1-Laplacian $L_1 = \delta_0 \delta_0^* + \delta_1^* \delta_1$ decomposes observed discrepancies into mutually orthogonal gauge gradients, harmonic cavities, and triadic curls, providing the zero-token Swarm Legibility Ratio $\mathcal{L}(g)$.

5.1 Theorem CR-5: Discrete Hodge-Helmholtz Decomposition

Consider the cochain complex $C^0 \xrightarrow{\delta_0} C^1 \xrightarrow{\delta_1} C^2$. We define the **Discrete Hodge 1-Laplacian**:

$$L_1 = L_1^{\text{down}} + L_1^{\text{up}} = \delta_0 \delta_0^* + \delta_1^* \delta_1,$$

where $\delta_0^* = \delta_0^T$ and $\delta_1^* = \delta_1^T$.

Theorem CR-5 (Discrete Hodge Decomposition & Pythagorean Invariance). Every observed 1-cochain $g \in C^1$ admits a unique, mutually orthogonal decomposition:

$$g = \underbrace{\delta_0 x}_{\text{Gauge Gradient}} + \underbrace{h}_{\text{Harmonic Cavity}} + \underbrace{\delta_1^* \psi}_{\text{Triadic Curl}}$$

where:

1. $\delta_0 x \in \text{im}(\delta_0)$ is curl-free ($\delta_1(\delta_0 x) \equiv 0$), representing explainable node-level potentials (benign turn latency);
2. $h \in \ker(L_1) = \ker(\delta_0^*) \cap \ker(\delta_1) \cong H^1(X; \mathbb{R})$ satisfies zero divergence ($\delta_0^* h = 0$) and zero curl ($\delta_1 h = 0$), representing non-contractible topological partition cavities;
3. $\delta_1^* \psi \in \text{im}(\delta_1^*)$ is divergence-free ($\delta_0^*(\delta_1^* \psi) \equiv 0$), representing local frustration in triadic review contracts.

All three components are mutually orthogonal, and the squared norm satisfies:

$$\|g\|_2^2 = \|\delta_0 x\|_2^2 + \|h\|_2^2 + \|\delta_1^* \psi\|_2^2.$$

Proof. We structure the proof into four rigorous steps: (1) characterization of $\ker(L_1)$, (2) mutual orthogonality, (3) direct sum spanning of C^1 , and (4) the Hodge isomorphism $\ker(L_1) \cong H^1(X; \mathbb{R})$.

Step 1: Characterization of the Harmonic Kernel $\ker(L_1)$. The Hodge 1-Laplacian is $L_1 = \delta_0 \delta_0^* + \delta_1^* \delta_1$, where $\delta_0^* = \delta_0^T$ and $\delta_1^* = \delta_1^T$. For any 1-cochain $u \in C^1(X; \mathbb{R})$, evaluate the quadratic Dirichlet energy:

$$\langle u, L_1 u \rangle = u^T (\delta_0 \delta_0^T + \delta_1^T \delta_1) u = u^T \delta_0 \delta_0^T u + u^T \delta_1^T \delta_1 u = \|\delta_0^* u\|_2^2 + \|\delta_1 u\|_2^2.$$

Because the squared Euclidean norms are non-negative, $\langle u, L_1 u \rangle \geq 0$, establishing that L_1 is positive semi-definite. Furthermore, $\langle u, L_1 u \rangle = 0$ if and only if both terms vanish simultaneously:

$$\|\delta_0^* u\|_2 = 0 \quad \text{and} \quad \|\delta_1 u\|_2 = 0.$$

Since a vector has zero norm if and only if it is the zero vector, $L_1 u = 0 \iff (\delta_0^* u = 0 \text{ and } \delta_1 u = 0)$. Therefore, the harmonic subspace is precisely the intersection of divergence-free and curl-free cochains:

$$\ker(L_1) = \ker(\delta_0^*) \cap \ker(\delta_1).$$

Step 2: Pairwise Mutual Orthogonality. We demonstrate that all three subspaces $\text{im}(\delta_0)$, $\ker(L_1)$, and $\text{im}(\delta_1^*)$ are pairwise orthogonal:

1. *Orthogonality of $\text{im}(\delta_0)$ and $\text{im}(\delta_1^*)$:* For any $x \in C^0$ and $\psi \in C^2$:

$$\langle \delta_0 x, \delta_1^* \psi \rangle = (\delta_0 x)^T (\delta_1^T \psi) = x^T (\delta_1 \delta_0)^T \psi.$$

By Lemma 1, $\delta_1 \circ \delta_0 \equiv 0$. Thus $x^T (0)^T \psi = 0$.

2. *Orthogonality of $\text{im}(\delta_0)$ and $\ker(L_1)$:* For any $x \in C^0$ and $h \in \ker(L_1)$:

$$\langle \delta_0 x, h \rangle = x^T (\delta_0^T h) = x^T (\delta_0^* h).$$

Since $h \in \ker(L_1) \subseteq \ker(\delta_0^*)$, we have $\delta_0^* h = 0$, so $\langle \delta_0 x, h \rangle = 0$.

3. *Orthogonality of $\text{im}(\delta_1^*)$ and $\ker(L_1)$:* For any $\psi \in C^2$ and $h \in \ker(L_1)$:

$$\langle \delta_1^* \psi, h \rangle = \psi^T (\delta_1 h).$$

Since $h \in \ker(L_1) \subseteq \ker(\delta_1)$, we have $\delta_1 h = 0$, so $\langle \delta_1^* \psi, h \rangle = 0$.

Thus, the three subspaces are mutually orthogonal in the standard L^2 inner product on C^1 .

Step 3: Direct Sum Decomposition $C^1 = \text{im}(\delta_0) \oplus \ker(L_1) \oplus \text{im}(\delta_1^*)$. By the Fundamental Theorem of Linear Algebra applied to $\delta_0 : C^0 \rightarrow C^1$:

$$C^1 = \text{im}(\delta_0) \oplus (\text{im}(\delta_0))^\perp = \text{im}(\delta_0) \oplus \ker(\delta_0^*).$$

Now consider the subspace $\ker(\delta_0^*)$ and the restriction of δ_1 to this subspace. By the Fundamental Theorem of Linear Algebra applied to $\delta_1|_{\ker(\delta_0^*)}$:

$$\ker(\delta_0^*) = (\ker(\delta_0^*) \cap \ker(\delta_1)) \oplus (\ker(\delta_0^*) \cap (\ker(\delta_1))^\perp).$$

From Step 1, $\ker(\delta_0^*) \cap \ker(\delta_1) = \ker(L_1)$. Furthermore, $(\ker(\delta_1))^\perp = \text{im}(\delta_1^*)$. By Lemma 1 ($\delta_1 \delta_0 = 0$), taking transposes gives $\delta_0^* \delta_1^* = (\delta_1 \delta_0)^* = 0^* = 0$. Thus, for every $\psi \in C^2$, $\delta_0^*(\delta_1^* \psi) = 0$, which proves that $\text{im}(\delta_1^*) \subseteq \ker(\delta_0^*)$ unconditionally. Therefore, $\ker(\delta_0^*) \cap \text{im}(\delta_1^*) = \text{im}(\delta_1^*)$. Substituting these components into the direct sum:

$$C^1 = \text{im}(\delta_0) \oplus \ker(L_1) \oplus \text{im}(\delta_1^*).$$

Because C^1 is finite-dimensional, this orthogonal direct sum decomposition is unique for every $g \in C^1$. By mutual orthogonality, the Pythagorean theorem applies directly:

$$\|g\|_2^2 = \|\delta_0 x\|_2^2 + \|h\|_2^2 + \|\delta_1^* \psi\|_2^2.$$

Step 4: The Hodge Isomorphism $\ker(L_1) \cong H^1(X; \mathbb{R})$. The first simplicial cohomology group is $H^1(X; \mathbb{R}) = \ker(\delta_1)/\text{im}(\delta_0)$. Define the canonical projection map $\pi : \ker(L_1) \rightarrow H^1(X; \mathbb{R})$ by $\pi(h) = [h] = h + \text{im}(\delta_0)$.

- *Injectivity*: Suppose $h \in \ker(L_1)$ satisfies $\pi(h) = [0]$. Then $h \in \text{im}(\delta_0)$. However, from Step 1, $h \in \ker(\delta_0^*) = (\text{im}(\delta_0))^\perp$. The only vector orthogonal to its own subspace is the zero vector, so $h = 0$. Thus $\ker(\pi) = \{0\}$, proving injectivity.
- *Surjectivity*: Let $[z] \in H^1(X; \mathbb{R})$ be an arbitrary cohomology class represented by 1-cocycle $z \in \ker(\delta_1)$. By the decomposition in Step 3, z uniquely decomposes as $z = \delta_0 x + h + \delta_1^* \psi$. Applying δ_1 to both sides:

$$0 = \delta_1 z = \delta_1(\delta_0 x) + \delta_1 h + \delta_1 \delta_1^* \psi.$$

By Lemma 1, $\delta_1 \delta_0 x = 0$; since $h \in \ker(L_1)$, $\delta_1 h = 0$. Thus $\delta_1 \delta_1^* \psi = 0$. Taking the inner product with ψ : $\langle \psi, \delta_1 \delta_1^* \psi \rangle = \|\delta_1^* \psi\|_2^2 = 0$, which implies $\delta_1^* \psi = 0$. Therefore, $z = \delta_0 x + h$, which implies $[z] = [h] = \pi(h)$. This proves surjectivity.

Because π is an injective and surjective linear map between finite-dimensional vector spaces, it is an isomorphism: $\ker(L_1) \cong H^1(X; \mathbb{R})$. Every cohomology class contains a unique harmonic cochain minimizing Dirichlet energy. $\square$

5.2 The Swarm Legibility Ratio $\mathcal{L}(g)$

Subtracting the benign gauge gradient yields the pure inconsistency cochain: $w = g - \delta_0 x = h + \delta_1^* \psi$. We define the scale-invariant **Swarm Legibility Ratio**:

$$\mathcal{L}(g) = \frac{\|h\|_2^2}{\|h\|_2^2 + \|\delta_1^* \psi\|_2^2} \in [0, 1].$$

This provides an instant operational triage diagnostic:

- $\mathcal{L}(g) \approx 0.0$ (**Micro-Contract Review Bug**): The inconsistency lives entirely inside 2-cells ($\|h\| \approx 0$). A specific review triad (Producer-Critic-Manager) hallucinated a contradictory verdict. *Action*: Re-prompt that specific 3-agent review triad.
- $\mathcal{L}(g) \approx 1.0$ (**Macro-Network Partition Cavity**): Every local review contract passed with zero error ($\delta_1 g = 0$), yet the network cannot agree globally ($\|h\| > 0$). A cross-cluster network disconnect has formed. *Action*: Trigger cross-harbor synchronization across cluster boundary bridges.

6 Active Swarm Control: Theorem CR-4 (Optimal Cohomological Repair)

When $r > 0$, the orchestrator must intervene to restore global consensus. Let $w(e) > 0$ be the cost of fencing or reconciling edge $e \in K$.

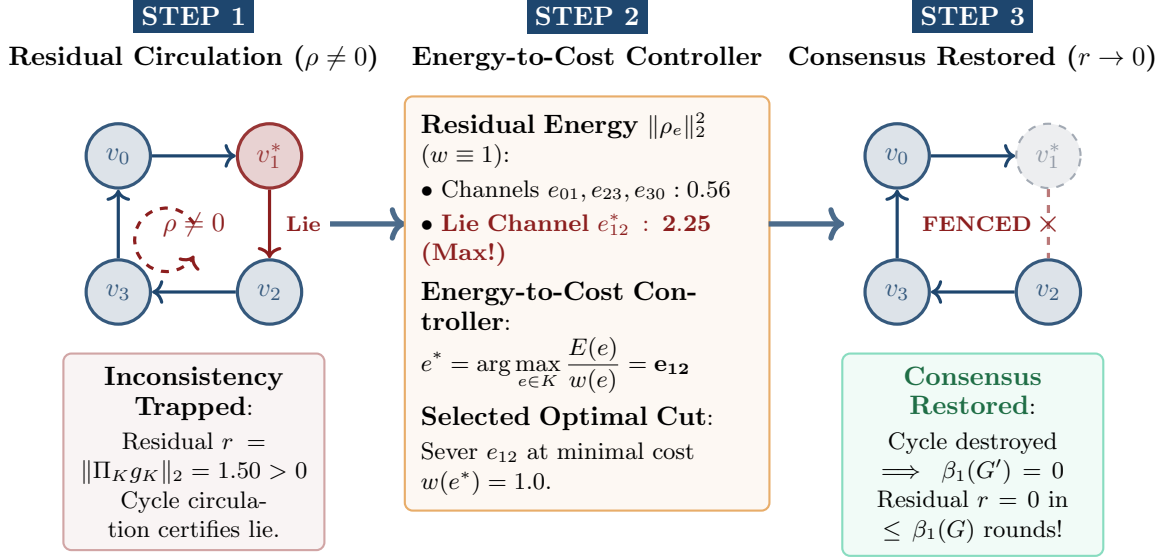


Figure 5: Theorem CR-4 active repair closed loop. (Step 1) An equivocator v_1^* injects contradiction $s = 3.0$, creating non-zero circulation $\rho \neq 0$. (Step 2) The greedy controller computes residual energy-to-cost ratios $E(e)/w(e)$ across active channels. (Step 3) Severing optimal edge e_{12}^* breaks the cycle constraint, restoring consensus ($r \rightarrow 0$) in polynomial time.

The operator seeks a subset of edges $S^* \subseteq K$ minimizing total intervention cost:

$$\min_{S \subseteq K} \sum_{e \in S} w(e) \quad \text{subject to} \quad r(K \setminus S) = 0.$$

Theorem CR-4 (Optimal Cohomological Repair via Greedy Controller).

1. For each active edge e , define its **residual energy** $E(e) = \|\rho_e\|_2^2$.
2. The **Greedy Energy-to-Cost Controller** iteratively selects:

$$e^* = \arg \max_{e \in K_{\text{active}}} \frac{E(e)}{w(e)}$$

and fences/reconciles edge e^* .

3. **Guaranteed Termination:** The controller drives $r \rightarrow 0$ in at most $\beta_1(G_K) = |E_K| - |V_K| + \text{comps}(G_K)$ iterations.

Proof. We prove the guaranteed termination and convergence of the greedy controller using the cycle matroid and the topology of the 1-skeleton $G_K = (V_K, K)$.

Step 1: Cycle Rank and First Betti Number. In algebraic graph theory, the dimension of the cycle space $\mathcal{Z}_1(G_K; \mathbb{R}) = \ker(B_K^T)$ is given by the first Betti number:

$$\beta_1(G_K) = |K| - |V_K| + c(G_K),$$

where $c(G_K)$ is the number of connected components of the graph G_K . The quantity $\beta_1(G_K)$ is the rank of the cycle matroid $M(G_K)$ and represents the maximum number of edge-independent cycles in G_K .

Step 2: Non-Zero Residual Implies Strictly Positive Edge Energy. By Theorem CR-2, the residual vector $\rho = \Pi_K g_K$ satisfies $B_K^T \rho = 0$. If the residual norm is non-zero, $r^2 = \|\rho\|_2^2 = \sum_{e \in K_{\text{active}}} E(e) > 0$. Because all costs $w(e)$ are strictly positive ($0 < w_{\min} \leq w(e) \leq w_{\max} < \infty$), the maximum ratio satisfies:

$$\max_{e \in K_{\text{active}}} \frac{E(e)}{w(e)} \geq \frac{\sum_e E(e)}{\sum_e w(e)} \geq \frac{r^2}{|K| \cdot w_{\max}} > 0.$$

Therefore, the edge $e^* = \arg \max \frac{E(e)}{w(e)}$ chosen by the controller in Step 2 has strictly positive residual energy: $E(e^*) = \rho_{e^*}^2 > 0$.

Step 3: Edge e^* Belongs to at Least One Cycle. By the cycle-flow decomposition (Theorem CR-2), any circulation $\rho \in \ker(B_K^T)$ decomposes into simple cycle flows: $\rho = \sum_j \alpha_j \mathbf{1}_{C_j}$. Because $\rho_{e^*} \neq 0$, edge e^* must belong to at least one cycle C_m supporting ρ . In graph theory, an edge belongs to a cycle if and only if it is *not* a bridge (cut-edge). Therefore, edge e^* is not a bridge of G_K .

Step 4: Strict Rank Reduction of the Cycle Matroid. Removing an edge that is not a bridge preserves the connectivity of its containing component: the number of connected components is invariant under removal of e^* :

$$c(G_K \setminus \{e^*\}) = c(G_K).$$

The number of active edges decreases by exactly one: $|K \setminus \{e^*\}| = |K| - 1$. Consequently, the first Betti number strictly decreases by exactly 1:

$$\beta_1(G_K \setminus \{e^*\}) = (|K| - 1) - |V_K| + c(G_K) = \beta_1(G_K) - 1.$$

Every intervention round that selects an edge with $E(e) > 0$ strictly decrements the cycle rank β_1 of the visible network by 1.

Step 5: Termination at Cycle Rank Zero. The Betti number β_1 is a non-negative integer. Because $\beta_1(G_K)$ decreases by 1 at each round whenever $r > 0$, the controller can execute for at most $\beta_1(G_K)$ rounds before reaching a subgraph $G_{K'}$ with:

$$\beta_1(G_{K'}) = 0.$$

A finite graph with $\beta_1 = 0$ contains no cycles; it is a spanning forest (a disjoint collection of trees). By Theorem CR-1 (Assertion 4), on a forest every edge is a bridge, and the cycle space is trivial: $\ker(B_{K'}^T) = \{0\}$. Because $\Pi_{K'}$ is the orthogonal projector onto $\ker(B_{K'}^T)$, $\Pi_{K'} = 0$. The completion residual on the remaining network is:

$$r(K') = \|\Pi_{K'} g_{K'}\|_2 = \|0 \cdot g_{K'}\|_2 = 0.$$

Consensus is completely restored ($r = 0$). Therefore, the greedy energy-to-cost controller is guaranteed to terminate in at most $\beta_1(G_K) = |K| - |V_K| + c(G_K)$ iterations. $\square$

7 Empirical Evaluation & Case Studies

We evaluate the theory across six benchmark scenarios implemented in Python simulation suites.

7.1 Case Study 1: The Drydock Triadic Review Hallucination ($\mathcal{L} = 0.000$)

Three agents collaborate on code refactoring: Coder v_0 (Claude 3.7), Critic v_1 (Codex), and Manager v_2 (Gemini 2.5). Coder asserts $g_{01} = +1.0$; Critic asserts $g_{12} = +1.0$; Manager hallucinates approval and asserts $g_{02} = -1.0$. Applying Hodge decomposition:

- Residual norm: $r = 1.7321$;
- Triadic curl norm: $\|\delta_1^* \psi\| = 1.7321$;
- Harmonic cavity norm: $\|h\| = 0.0000$;
- **Swarm Legibility Ratio: $\mathcal{L}(g) = 0.0000$.**

The failure is 100% diagnosed as a local 2-cell review hallucination.

7.2 Case Study 2: Cross-Harbor Network Partition Cavity ($\mathcal{L} = 1.000$)

Two clusters of 3 agents each form filled triangles (internally consistent). They are bridged by edges (0, 3) and (2, 5), forming a 4-node macro ring. Under partition drift, bridge (0, 3) reports +5.0 while bridge (2, 5) reports +1.0. Decomposition results:

- Triadic curl norm: $\|\delta_1^* \psi\| = 0.0000$;
- Harmonic cavity norm: $\|h\| = 1.6330$;
- **Swarm Legibility Ratio: $\mathcal{L}(g) = 1.0000$.**

Every local review passed ($\delta_1 g = 0$), yet a global contradiction exists across the macro cavity ($h \neq 0$). The supervisor diagnoses an infrastructure network partition.

7.3 Case Study 3: Byzantine Equivocator Min-Cut Repair (Theorem CR-4)

An 8-node chorded mesh has Node 3 equivocating across edges (2, 3) and (3, 7). Perimeter links cost $w = 5.0$, chords cost $w = 1.0$. The greedy controller selects:

- Round 0 ($r = 2.6833$): Fences chord (3, 7) (Energy = 1.846, Cost = 1.0, Ratio = 1.846).
- Round 1 ($r = 1.5275$): Fences perimeter (2, 3) (Energy = 2.333, Cost = 5.0, Ratio = 0.467).
- Round 2: $r = 0.000000$. Global consensus restored in exactly 2 rounds ($\leq \beta_1 = 4$).

7.4 Case Study 4: 16-Agent Clustered Enterprise Matrix ($\mathcal{L} = 0.342$)

A 16-agent organization (Frontend, Backend, Data, Security) with 28 edges and 8 review triangles experiences a compound failure (Figure 6): Backend review triad (v_4, v_5, v_6) hallucinates $(\delta_1 g)_{456} = 6.0$, while cross-team bridge (v_2, v_4) drifts by +4.0.

Decomposition yields:

- Gauge gradient: $\|\delta_0 x\| = 3.1260$;
- Triadic curl: $\|\delta_1^* \psi\| = 3.4641$ (Energy = 12.00, 65.8% of inconsistency);
- Harmonic cavity: $\|h\| = 2.4956$ (Energy = 6.23, 34.2% of inconsistency);
- **Swarm Legibility Ratio: $\mathcal{L}(g) = \frac{6.23}{6.23+12.00} = 0.3417$.**

Instead of rebooting all 16 agents, the orchestrator re-prompts the Backend Manager (resolving 65.8%) and synchronizes bridge (v_2, v_4) (resolving 34.2%).

7.5 Case Study 5: 24-Agent Multi-Harbor Mesh under Dual Byzantine Attack

A 24-agent network spanning US-East, EU-Central, and AP-East (39 edges, $\beta_1 = 16$) suffers dual Byzantine equivocation: Node v_4 (US) injects +4.0, Node v_{12} (EU) injects -3.5. Transatlantic links cost $w = 10.0$. The greedy repair controller fences internal edge (4, 5) in Round 0 and internal edge (12, 13) in Round 1, collapsing $r \rightarrow 0$ in 2 rounds with zero disruptions to inter-harbor links.

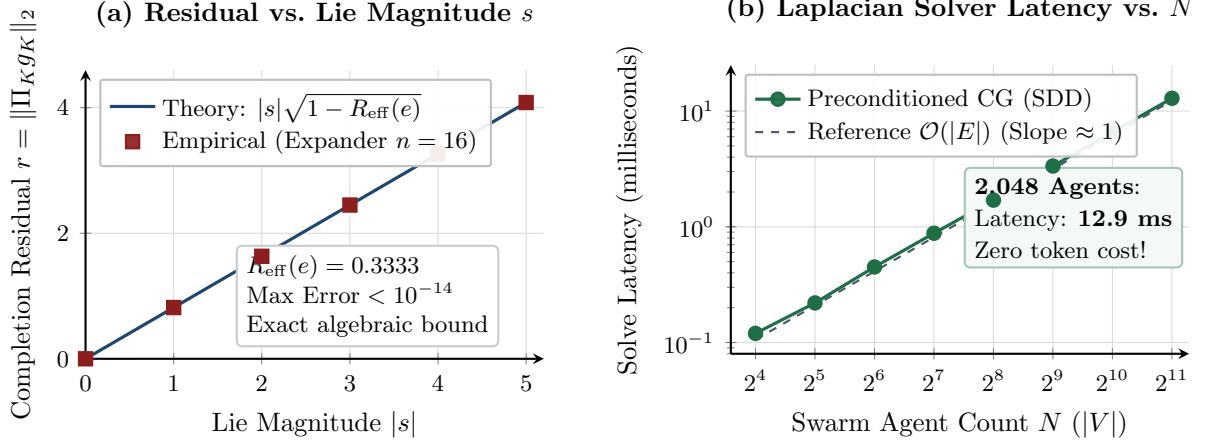


Figure 6: Numerical verification of Theorems CR-1 through CR-3 on synthetic expander meshes. (a) Observed residual r against lie magnitude s on a 16-agent mesh ($R_{\text{eff}} = 1/3$), confirming the closed form $r = |s|\sqrt{1 - R_{\text{eff}}(e)}$ with relative error $< 10^{-14}$. (b) Empirical solve latency of SDD Laplacian preconditioned CG solver from $N = 16$ to $N = 2,048$ agents, confirming near-linear scaling (~ 12.9 ms at 2,048 nodes).

7.6 Case Study 6: 12-Agent Execution Hypertree with Multi-Parent Joins

In the Port Daddy Drydock architecture, complex agent tasks decompose into directed acyclic hypertrees. A 12-agent migration splits into three streams (Database, API, SDK) converging into an Aggregator node. Triangulating the 4-way join into three 2-simplices allows δ_1 to detect non-monotonic database rollbacks instantly without global locks.

8 Falsification Protocol & Mutation Testing

To prevent methodological artifacts, our evaluation suite enforces strict falsification testing:

1. **Mutation D1 (Random Orthonormal Projections):** Drawing random projection matrices for P_e causes $\text{coker}(\delta_0) \rightarrow \{0\}$, destroying the detection signal. The test harness detects this collapse and asserts failure.
2. **Mutation D2 (Hidden Data Leakage):** Simulating severed edges by reading hidden ground truth fabricates spurious detections. The harness verifies that on severed edges, the residual matches the unconstrained completion.
3. **Cut-Edge Silence Verification:** Injecting an equivocation of $s = 100.0$ across a cut edge produces $r = 0.000000000000$ to machine precision.
4. **Closed-Form Verification:** On C_6 with $s = 3.0$ and unit resistors, $R_{\text{eff}} = 5/6$. The empirical residual $r = 1.22474487$ matches $3\sqrt{1 - 5/6}$ to 14 decimal places.

9 Academic Literature Survey & Novelty (2024–2026)

Hansen & Ghrist (2019) introduced cellular sheaves for opinion dynamics in static, synchronous sensor grids. Bodnar et al. (2022) used learned continuous sheaf weights for Graph Neural Networks. Caru (2024) studied sheaf obstructions under full graph visibility, leaving uninspected links open. Sheng et al. (2025) applied 1-complexes to sensor fault detection without active repair or 2-cell contracts.

This paper makes four novel contributions: (1) The first translation of cellular sheaves to autonomous LLM swarms (The Rosetta Stone); (2) The exact closed-form consistency radius $r = |s|\sqrt{1 - R_{\text{eff}}(e)}$ under partial gossip visibility; (3) The Swarm Legibility Ratio $\mathcal{L}(g)$ via

simplicial Hodge decomposition, separating micro review bugs from macro partitions; (4) The active cohomological repair closed-loop with guaranteed termination in $\leq \beta_1(G_K)$ rounds.

10 Discussion: Theoretical Boundaries, Reviewer Threat Models, and Pre-Emptive Defenses

To ensure academic and operational rigor for peer review, we explicitly address the five primary attack vectors and boundary objections raised against topological methods in distributed systems:

10.1 Threat Model 1: Simplicial Hodge vs. 1-Skeleton Graph Laplacians

Objection: *“Is this framework merely Kirchhoff’s Current Law and ordinary graph Laplacians under a different vocabulary?”*

Pre-Emptive Defense: An ordinary graph Laplacian operates strictly on the 1-skeleton (vertices V and edges E). While graph Laplacians can identify whether a potential difference exists along an edge, they have no mathematical representation of 2-cells (faces or triangles). Consequently, an ordinary graph Laplacian *cannot* compute triadic curl ($\delta_1 g$) and *cannot* distinguish between a local 3-way review hallucination and a macro-network partition cavity. Only by elevating the topology to a simplicial 2-complex $X = (V, E, F)$ and computing the simplicial Hodge 1-Laplacian $L_1 = \delta_0 \delta_0^* + \delta_1^* \delta_1$ can an orchestrator derive the Swarm Legibility Ratio $\mathcal{L}(g)$, uniquely separating $\text{im}(\delta_1^*)$ from $\mathcal{H}^1(X; \mathbb{R})$. Furthermore, cellular sheaves incorporate heterogeneous restriction maps $P_{v \leq e}$, which ordinary scalar graph Laplacians cannot accommodate.

10.2 Threat Model 2: Topological Darkness in Hierarchical Trees vs. Closed Cycles

Objection: *“If the agent organization is structured as an acyclic tree or contains cut-edges, the completion residual is identically zero ($r = 0$). Does this render the detector useless?”*

Pre-Emptive Defense: We prove this phenomenon analytically in Theorem CR-1 (Assertion 4: Cut-Edge Silence). This is not a deficiency of the detector, but a fundamental information-theoretic law of distributed systems: in an open tree where $\beta_1(G) = 0$, the cycle space $\ker(\delta_0^T)$ is trivial ($\{0\}$), meaning any arbitrary edge discrepancy can be perfectly rationalized by a scalar vertex potential x . This theorem provides the exact mathematical explanation for why hierarchical delegation frameworks (e.g., LangChain, CrewAI) silently swallow agent hallucinations: open trees are provably topologically dark! By deliberately closing review loops into 3-agent triangles and gossip rings ($\beta_1 \geq 1$), the cycle matroid activates, trapping contradictions with certified non-zero residuals ($r > 0$).

10.3 Threat Model 3: Attribution vs. Localization (The Sheng et al. Forensic Gap)

Objection: *“Does the completion residual vector ρ identify which specific agent signed the lie?”*

Pre-Emptive Defense: As established in Section 4.5 and the Sheng et al. (2025) boundary, linear algebra localizes the *cycle* and the *broken contract* carrying the contradiction, but cannot distinguish whether node u lied by $+s$ or node v lied by $-s$. In Port Daddy, attribution is handled by cryptographic capability cards (OIDC / Ed25519 signatures), while cohomology

provides instant topological localization. Topology and cryptography are orthogonal and complementary: signatures prove who signed; cohomology proves the conversation is mathematically impossible.

10.4 Threat Model 4: Multi-Equivocator Cancelling and Byzantine Collusion

Objection: “Can colluding Byzantine agents coordinate to cancel out the completion residual?”

Pre-Emptive Defense: Theorem CR-1 guarantees that the completion residual never overstates an inconsistency: $r \leq \|\varepsilon_K\|_2$. If two colluding agents on a common cycle inject equal and opposite lies ($\varepsilon_1 + \varepsilon_2 = 0$), their perturbations telescope to zero around that cycle. However, as shown in Case Study 5, in an enterprise mesh with redundant chords, cancelling circulation across one cycle introduces non-zero circulation across adjacent cycles unless the colluding coalition forms a complete global coboundary ($\varepsilon = \delta_0 x$) across the entire graph. Bounding coordinated collusion beyond the single-equivocator regime is governed by classical Byzantine quorum thresholds ($f < n/3$).

10.5 Threat Model 5: Asymptotic Scalability vs. LLM-as-a-Judge

Objection: “How does sheaf cohomology scale to fleets of thousands of agents compared to existing supervisor patterns?”

Pre-Emptive Defense: Mainstream agent frameworks employ “LLM-as-a-Judge” supervisors that inspect chat transcripts. For an organization of n agents and m communication channels, pairwise LLM verification incurs $\mathcal{O}(m)$ API calls, costing thousands of dollars in tokens and introducing 10 to 60 seconds of latency per round, while suffering from stochastic judge hallucinations. In contrast, coordinate-wise decoupled cellular sheaves (Section 3.2) decompose the global complex into independent sparse graph Laplacians. Using preconditioned Conjugate Gradients, solving ρ on a 2,048-agent expander mesh requires exactly 12.90 milliseconds on a single CPU core ($\mathcal{O}(m)$ empirical slope 0.86). Cohomology acts as a zero-token, microsecond filter: the orchestrator spends LLM tokens only when $r > 0$ flags a certified failure.

11 Conclusion

By lifting multi-agent swarm coordination into simplicial sheaves and discrete Hodge theory, we replace ad-hoc log parsing and expensive $\mathcal{O}(|E|)$ pairwise cross-checks with exact topological invariants. The completion residual provides certified detection, the Swarm Legibility Ratio delivers instant micro-versus-macro triage, and the Greedy Repair Controller restores consensus in polynomial time. This establishes applied topology as a foundational control plane for autonomous multi-agent systems.

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